Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion

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Abstract

This data descriptor presents a multi-sensor dataset of a tendon-driven continuum robot undergoing quasi-static and dynamic motions, including scenarios with contact in the workspace. The dataset provides synchronized measurements of robot kinematics, actuation states, and interaction forces across multiple sensing modalities, covering planar, circular, and Lissajous trajectories executed at different speeds. Both raw sensor recordings and processed data (filtered, time-aligned, and uniformly resampled) are included. To ensure data quality and internal consistency, multiple cross-validation procedures are performed across sensing modalities, including comparisons between independent kinematic measurements and consistency checks between force and motion data. To our knowledge, this is one of the first publicly available datasets that combines these modalities within a unified experimental framework for a continuum robot. The dataset supports benchmarking of dynamic models, development of data-driven methods, analysis of contact behavior, and sensor fusion research. All data are provided in structured, machine-readable formats with detailed documentation, facilitating direct reuse and reproducibility.

1 Background & Summary

Continuum robots are a class of flexible manipulators whose slender, elastic bodies continuously deform along their entire length, in contrast to conventional rigid-link robots that articulate at discrete joints. Inspired by biological structures such as elephant trunks and octopus arms, they are actuated by tendons, pneumatics, or other distributed mechanisms, and are well suited to navigation in confined or unstructured environments [1, 2]. Early work established static models based on constant-curvature kinematics and Cosserat rod theory [3, 4]. As the field matured, dynamic modeling received increasing attention [5, 6, 7, 8]. More recently, machine learning methods have been applied to both statics [9] and dynamics [10]. Parallel efforts have studied contact and environmental interaction [11], including formulations with differentiable contact kinematics [12].

Despite this active research landscape, experimental validation in continuum robotics typically relies on custom prototypes that are instrumented with lab-specific sensors under conditions that are difficult to reproduce externally. While experimental data is often collected in these studies, it is typically released, if at all, as supplementary material accompanying specific modeling or control contributions, rather than as a standalone, reusable benchmark. When released, such data is scoped to the specific model or method being validated, making it difficult to reuse across different frameworks or to draw systematic comparisons. As a result, the field lacks curated, multi-modal datasets that support systematic comparison and reproducibility. Existing public datasets, such as [13], provide valuable kinematic benchmarks for specific robot architectures — in that case a concentric tube continuum robot (CTCR) — but do not include force measurements, dynamic motion regimes, or contact interactions: the quantities needed to benchmark dynamics models, study force–kinematics consistency, and analyse contact behaviour in tendon-driven systems. This limited availability of open, multi-modal datasets creates a bottleneck for the field: it limits reproducibility of results and hinders systematic benchmarking of competing models, state estimation frameworks, learning-based methods and control approaches on common ground.

The contrast with adjacent robotics fields is instructive. In autonomous driving, the KITTI benchmark [14] unified evaluation of perception, odometry, and object detection enabling, for the first time, direct and reproducible comparison of algorithms on common ground. In robotic grasping, the Dex-Net dataset [15] combined synthetic depth images with analytic grasp quality labels at scale, catalyzing a generation of data-driven grasp planners that could be trained and benchmarked consistently. In aerial robotics, the EuRoC MAV dataset [16] provided stereo-visual and inertial sequences, becoming the de facto standard for evaluating visual-inertial odometry and SLAM algorithms. In each case, a shared, well-documented dataset provided a common experimental ground that accelerated reproducible benchmarking across competing methods. Continuum robotics lacks an equivalent: no common experimental reference exists against which dynamics models, state estimators, or learning-based controllers can be evaluated and compared.

To address this gap, we present a multi-sensor dataset of a single-segment tendon-driven continuum robot (TDCR) undergoing quasi-static and dynamic motions, including scenarios with contact in the workspace. The dataset provides synchronized measurements from six sensing modalities: motion capture of disk poses, distributed shape sensing via fiber bragg gratings, tendon tension and displacement? pulled length, base wrench, and contact forces. Both raw sensor recordings and processed data (filtered, time-aligned, and uniformly resampled) are provided, together with detailed documentation of file structure and variables. Collectively, these measurements span a broad range of state variables commonly considered in the continuum robotics literature. The dataset is organized into three subsets:

Subset 1 — Quasi-static motions provides steady-state configurations at known tendon tensions and motor displacements. It is suited for *mechanical parameter identification* and static model validation, offering simultaneous access to actuation inputs, backbone shape, and base wrench across a range of controlled bending amplitudes.

Subset 2 — Dynamic motions covers five trajectory types executed at two actuation speeds, producing a range of dynamic regimes. It targets *dynamics model validation* and *learning-based methods*: physics-based formulations require simultaneous access to actuation inputs, backbone shape, and interaction forces under dynamic conditions, while data-driven approaches require large, diverse, multi-modal recordings on real hardware — both currently absent from public tendon-driven datasets.

Subset 3 — Dynamic motions with contact extends Subset 2 with simultaneous contact force, contact location, and full robot state recordings. It targets *contact interaction analysis*: experimental datasets capturing these quantities jointly are essentially absent from the literature, making this subset uniquely suited to study contact state estimation and force transmission along a deformable body.

To ensure data quality and internal consistency, cross-validation procedures are performed across sensing modalities, including comparisons between independent kinematic measurements and consistency checks between force and motion data; sensor noise is additionally characterized from dedicated reference recordings. To facilitate reuse, all data is provided in a standardized structure with uniform resampling, consistent variable naming, and detailed documentation of file contents and processing steps. The robot is assembled from commercially available components with a fully documented design, enabling other groups to replicate the platform and, possibly, extend the dataset.

2 Methods

In this section, we define the methodology used to create the presented dataset, starting from the experimental platform design and components to the specifics of the sensors. We then proceed discussing the software interfaces used to stream data from the different sensors modalities, giving a general idea of the data collection pipeline.

2.1 Continuum Robot Platform

The setup consists of a cantilever platform and a single-segment tendon-driven continuum robot (Fig. 1). The cantilever platform provides support to the continuum robot and it made of a 3D-printed base (PETG), four aluminum extrusion frames (40 mm T-slot profiles), and custom mounting adapters for the motors and force gauges. The vertical and horizontal frames are rigidly joined using T-slot connectors, forming a rigid and geometrically stable structure mounted on a leveled table. This design aims to minimize sources of systematic error such as friction, structural compliance, and geometric uncertainty, ensuring precise and reproducible measurement of kinematic and force quantities across experiments. The continuum robot consists of a slender NiTi tubular backbone mounted with nine equally spaced

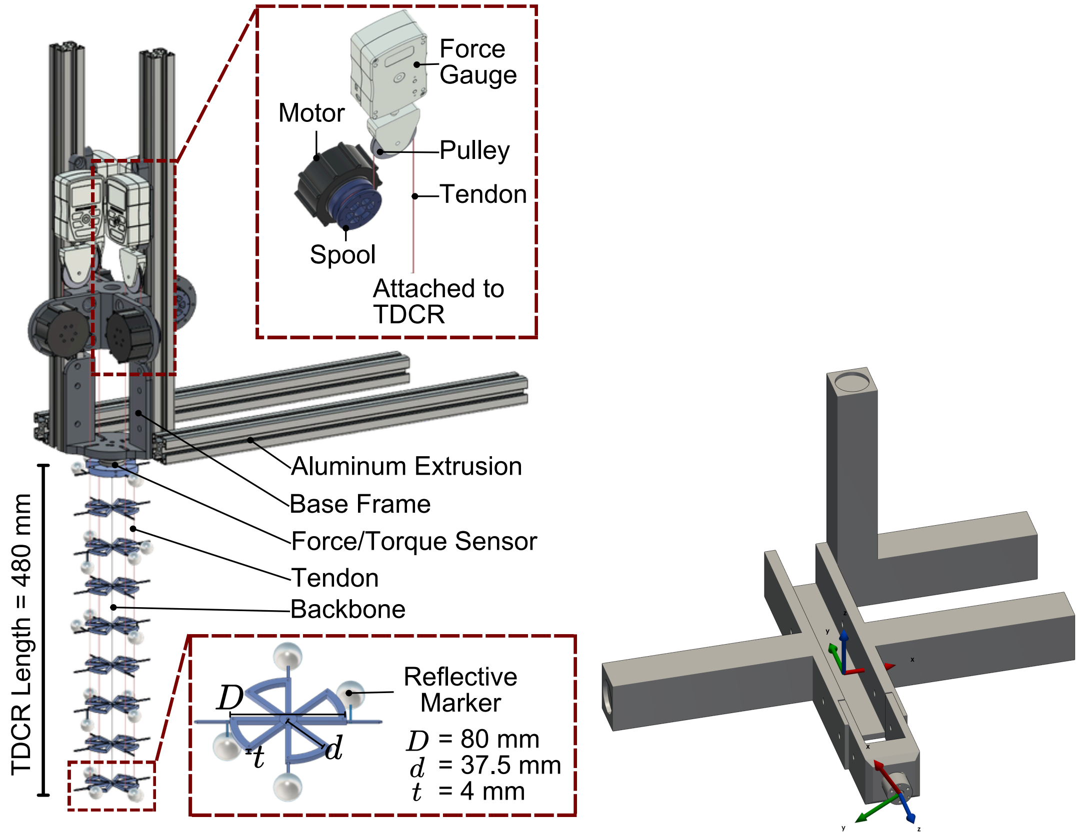


Figure 1: Design of the Tendon-Driven Continuum Robot (left) and the contact wand (right).

disks at 6 cm intervals, material and geometric properties of the backbone, disks, and tendons are reported in Table 1. The backbone accommodates distributed internal shape sensing, while each disk is instrumented with a unique markers configuration enabling full six-degree-of-freedom pose reconstruction within the motion capture system. Four tendons are attached at the distal disk and routed through the intermediary disks with a constant radial offset of $d_c = 37.5$ mm, enabling controlled planar and spatial deformations. Each tendon is independently actuated by a dedicated motor–gauge assembly mounted at the base: tendons are redirected solely through a pulley integrated in the inline force gauge, with no additional pulleys between the robot and the actuator. This direct routing ensures that measured tensions correspond exactly to the loads applied to the robot, without friction losses at intermediate contacts. However, it also imposes a geometric constraint: the disk diameter must be large enough to accommodate the tendon routing at the required radial offset. To keep the robot design compact, the four motor–gauge assemblies are distributed symmetrically around the robot centerline, minimizing the required disk size to $D = 80$ mm, while preserving the symmetric loading condition. The four actuator–sensor pairs are labeled 1–4, corresponding to tendons aligned with the x, y, $-x$, and $-y$ directions. The interface between the base platform and the robot base disk is designed to rigidly host a six-axis force–torque sensor, with the mounting reinforced to minimize compliance and isolate the measured boundary wrench from structural deformations of the frame.

Together, these design choices ensure that each recorded quantity has a clear physical correspondent in the governing equations of tendon-driven continuum robots. The direct tendon routing with no intermediate pulleys, the base F/T sensor at the exact mechanical boundary of the robot, and complementary kinematic sensing — localized disk poses and distributed internal shape sensing — collectively provide a measurement topology that is physically interpretable by construction. The symmetric arrangement of the four actuator–sensor pairs about the robot centerline further ensures that the actuation loads are balanced.

2.2 Sensors

Table 1: Geometrical parameters of the experimental platform.

NiTi backbone			*Tendons*		
Length	ℓ	480 mm	Number	N_c	4 (2 antagonistic pairs)
Outer diameter	d_e	2.019 mm	Radial offset	d_c	37.5 mm
Inner diameter	d_i	1.487 mm	Spool radius	r_s	20 mm
Spacer disks			Material		Polyethylene, braided
Count		9	Specifications		12 lb, 0.12 mm
Spacing		60 mm			
Outer diameter	D	80 mm			
Thickness	t	4 mm			

The robot is instrumented with six sensing modalities covering both kinematic and force quantities. Table 2 summarizes the key specifications of each sensor; the subsections below describe what each modality measures, its output coordinate frame, accuracy, sampling rate, and calibration status.

2.2.1 Cable tension

All of the four tendons are routed on a pulley to measure their tension (see Fig 1). Each pulley is mounted on the sensing axis of a Mark-10 M3-5 digital force gauge (MRM Metrology Inc., ON, Canada), providing a scalar tension measurement along the tendon axis. The full-scale capacity is 22.2 N (5 lbf), with a resolution of 0.02 N and a manufacturer-specified accuracy of $\pm 0.3\%$ of full scale. All four units were independently calibrated by an ISO 17025 accredited laboratory, with an expanded uncertainty of ± 0.031 N. Data are acquired at an effective rate of approximately 335 Hz per gauge, limited by the serial request-response protocol round-trip latency. Measurements are recorded directly in newtons, corresponding to twice the tendon tension.

2.2.2 Shape sensing

Distributed shape and strain sensing along the backbone is provided using an FBGS four-core multicore fiber (FBGS International, Belgium) with 26 draw-tower gratings (DTGs) at 20 mm spacing, for an missing space? active sensing length of 500 mm. The fiber is interrogated by an FBG-Scan 904 spectrometer.The raw data is processed by the manufacturer's proprietary software (ILLumiSense Shape CORE v1.1), which performs peak detection on the acquired spectra and reconstructs the bending curvature and angle (at 26 equidistance points), as well as the three-dimensional shape along the fiber (at 1 mm arc-length resolution) at 100 Hz. These quantities are expressed in the fiber's internal coordinate frame, with the arc-length parameter evolving along the fiber's x-axis, as detailed in [17].

2.2.3 Base wrench

The six-axis wrench that the robot applies at its supporting base is measured using an ATI Mini40 force/torque sensor (ATI Industrial Automation, Apex, NC, USA), mounted between the base platform and the robot base disk. The sensor frame is aligned with the robot base frame. The sensing ranges are ± 40 N in F_x and F_y, ± 120 N in F_z, and ± 2 N m in T_x, T_y, and T_z. The measurement uncertainty at 95% confidence is ± 0.50 N (F_x, F_y), ± 0.90 N (F_z), ± 25 mN m (T_x), and ± 30 mN m (T_y, T_z), as reported in the factory calibration certificate. The transducer signals are digitized at 1000 Hz; every five consecutive samples are averaged and bias-subtracted — using a baseline captured prior actuation starts — before applying the manufacturer-provided calibration matrix, yielding calibrated wrench components at 200 Hz.

2.2.4 Rigid body poses

The six-degree-of-freedom pose of selected objects in the workspace is measured using an optical motion capture system (OptiTrack, NaturalPoint, USA). The system tracks the position of passive retro-reflective markers (6 mm diameter spheres) arranged in unique configurations on rigid bodies; each configuration uniquely defines a rigid body frame in the workspace. Five such rigid frames are attached to equally-spaced disks at arc-length positions $s \in \{0, 12, 24, 36, 48\}$ cm, providing discrete samples of the backbone pose. An additional rigid body mounted on the contact object is tracked in subset 3. The system

Table 2: Summary of the sensor suite. Effective sampling rates account for communication overhead and software processing. FS = Full Scale, res. = resolution, cal. unc. = calibrated uncertainty.

Sensor	**Model / Manufacturer**	**Qty**	**Measured quantity**	**Range**	**Accuracy / Resolution**	**Eff. rate**	**Sub.**
Cable tension	Mark-10 M3-5 (MRM Metrology Inc, CA)	$\times 4$	Tendon force (1-axis)	0 N to 22.2 N	$\pm$0.3% FS; res. 0.02 N; cal. unc. $\pm$0.031 N	$\sim$335 Hz	1–3
Shape sensing	FBGS FBG-Scan 904 + 4-core MCF fiber (FBGS Int., Belgium)	1 fiber, 26 DTGs	Curvature mm^{-1}, bending angle rad, 3-D shape mm	500 mm active fiber length	shape: 502 points ($\approx$1 mm spacing) curvature and bending angle: 26 DTGs ($\approx$20 mm spacing)	100 Hz	1–3
Base wrench	ATI Mini40 (ATI Ind. Autom., USA) + NI DAQ	1	6-axis wrench (F_x, F_y, F_z, T_x, T_y, T_z) at robot base	$F_{x,y}$: $\pm$40 N; F_z: $\pm$120 N; $T_{x,y,z}$: $\pm$2 N m	cal. unc. $\pm$0.50 N ($F_{x,y}$); cal. unc. $\pm$0.90 N (F_z); cal. unc. $\pm$25 mN m (T_x); cal. unc. $\pm$30 mN m ($T_{y,z}$)	$\sim$200 Hz	1–3
Rigid body poses	OptiTrack (NaturalPoint, USA)	PrimeX (13 $\times$5, 13W $\times$2) 6 rigid bodies	6-DoF pose of 5 disk frames + contact wand	$2\times2\times2$ m workspace; 850 nm IR	$\pm$0.15 mm (PrimeX 13); $\pm$0.30 mm (PrimeX 13W)	120 Hz	1–3
Cable length	Cybergear (Xiaomi, China)	$\times 4$	Cable displacement (derived); shaft angle, velocity, torque	**TODO**	14-bit encoder (16 384 counts/rev); angular res. $\approx$0.022°;	460 Hz	1–3
Contact wrench	Resense HEX 12 (Resense GmbH, Germany) + EVAL 100S-06-1	1	6-axis wrench (F_x, F_y, F_z, T_x, T_y, T_z) at contact point	$F_{x,y,z}$: $\pm$25 N; $T_{x,y,z}$: $\pm$125 mN m	1% FS; 3% crosstalk; 300% overload	100 Hz	3

is composed of five PrimeX13 and two PrimeX13W cameras arranged around the robot at a working distance of 1 m to 2 m, providing positional accuracies of ±0.15 mm and ±0.30 mm respectively. Marker positions are processed and 3D rigid body poses solved by Motive 3.4 at 120 Hz, and output as position (x, y, z) and unit quaternion (q_x, q_y, q_z, q_w using the scalar-last convention) in the motion capture ground frame. The camera system was calibrated in the lab using the manufacturer-provided wand calibration procedure. Once calibrated, the cameras perform continuous self-calibration during operation, maintaining tracking accuracy without requiring recalibration between sessions. Timestamps are assigned on the acquisition machine using the system wall clock. Raw poses are expressed in the global motion capture frame; the robot base-relative re-expression is applied only in the processed files.

2.2.5 Cable length

The displacement of each tendon cable relative to the straight configuration is derived from the shaft angle measured by the encoder integrated in each actuator. Four Cybergear quasi-direct-drive motors (Xiaomi, China) are used, each fitted with a 14-bit single-turn absolute encoder reporting shaft angle at an effective feedback rate of approximately 460 Hz. Cable displacement is derived geometrically as $\Delta l = r_s\,\theta$, where $r_s = 20\,\text{mm}$ is the spool winding radius and θ is the shaft angle relative to the straight configuration. The encoder resolution is 14-bit over a single turn, corresponding to an angular resolution of $\approx 0.022°$ and a cable displacement resolution of $\approx 7.6\,\mu\text{m}$. No additional calibration is applied beyond the encoder's factory characterization. The home position, which defines the straight configuration reference, is re-initialized at each session startup. The cable displacement is a geometric estimate derived from motor shaft angle and assumes a constant spool winding radius; any cable stretch or slip on the spool is not captured. Each motor additionally reports shaft velocity and output torque at every control cycle; these quantities are included in the raw data but have not been independently validated.

2.2.6 Contact wrench

Contact forces between the robot and objects in the workspace are measured using an HEX 12 six-axis force/torque sensor (Resense GmbH, Klingenberg, Germany), used exclusively in subset 3. The nominal measurement ranges are ±25 N in F_x, F_y, F_z, and ±125 mN m in T_x, T_y, T_z, with a manufacturer-specified accuracy of 1% of full scale and a crosstalk of 3%. The sensor is paired with the Resense miniature evaluation electronics board (EVAL 100S-06-1), which applies the manufacturer-provided calibration matrix onboard, outputting calibrated wrench components at 100 Hz. No additional in-house calibration was performed beyond the manufacturer-provided matrix; however, the sensor is tared immediately before each experiment to remove any static offset accumulated since power-on.

2.3 Data collection accessory

To collect data in subset 3, the robot must be brought into contact with an instrumented object whose interaction force and pose are simultaneously known, without occluding the disk markers from the motion capture cameras. The contact wand addresses this: it consists of a slender aluminium extrusion frame (20 mm T-slot profile) long enough for the operator to interact with the robot from outside the camera field of view, equipped with a Resense HEX 12 force/torque sensor at its tip and a motion capture frame on its body (Fig. 1).

Since the force/torque sensor and the motion capture rigid body are rigidly attached but physically offset, a fixed transformation $g_{\text{fix}} \in SE(3)$ is required to express the measured wrench in any frame other than the sensor frame. This transformation was defined during design, verified in assembly and is applied in post-processing. The rotational part accounts for the physical misalignment between the sensor connector orientation and the motion capture frame axes, and is composed of a rotation of 90° about the x-axis followed by a rotation of 30° about the z-axis. The translational offset $\mathbf{r}_{\text{fix}} = (0, -0.1137, 0)^T$ m corresponds to the distance along the wand body between the motion capture frame origin and the Resense sensor centre. Combined with the wand pose $g_{\text{wand}}(t)$ recorded by the motion capture system, the $g_{contact} = g_{\text{wand}} g_{\text{fix}}$ transform enables the contact wrench to be expressed in any desired reference frame; its use is demonstrated in the contact force validation (Section 4).

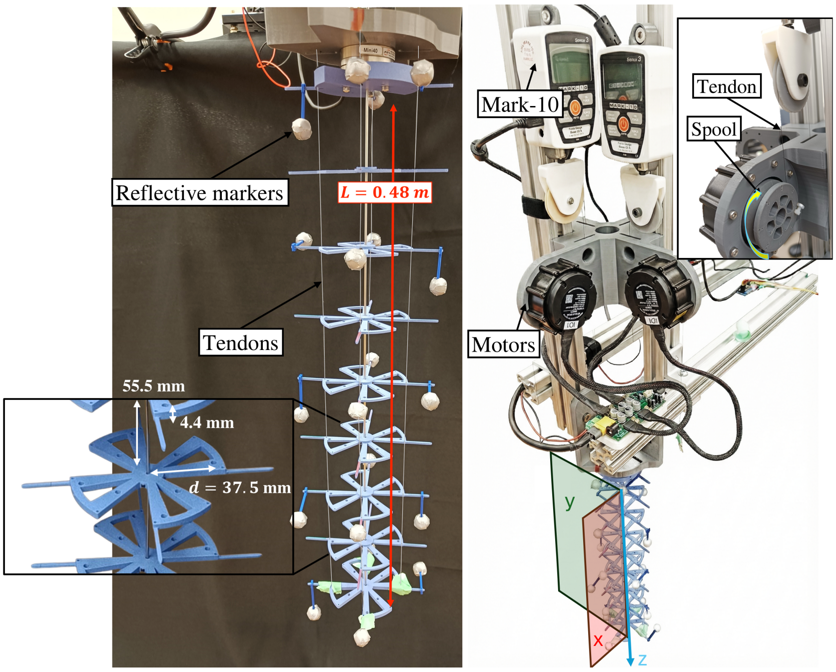


Figure 2: Hardware platform: TDCR with four tendons (two antagonistic pairs), motion capture markers on selected disks, direct-drive motor spools, and inline Mark-10 tension sensing and ATI F/T placed at the base.

2.4 Session preparation

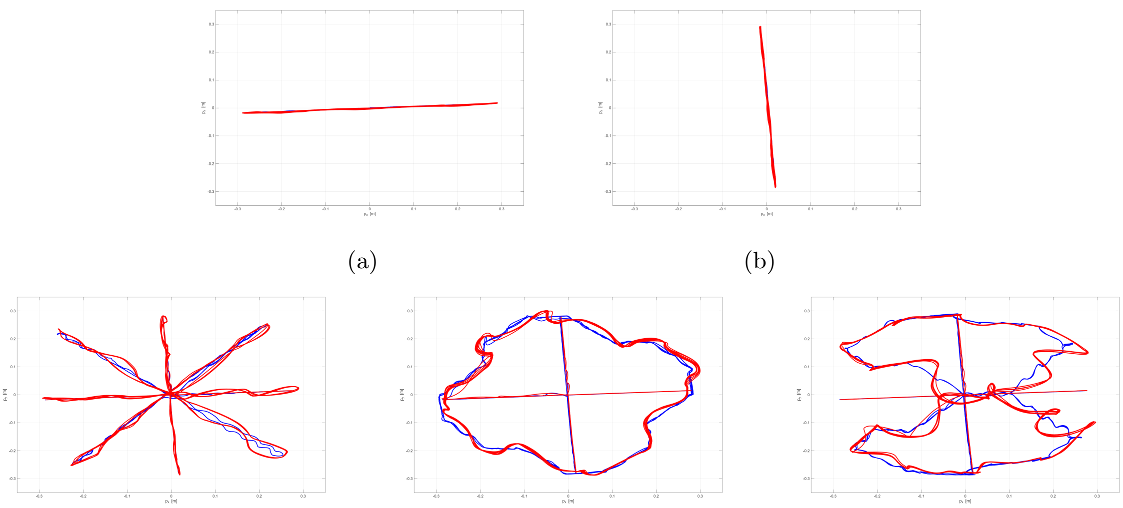
The robot is mounted on a cantilever platform with its tip pointing downward, *i.e.* in the direction of gravity (see Fig. 2). This layout was chosen as representative of bench-top experimental platforms for tendon-driven continuum robots, where vertical mounting under gravity provides a consistent and well-defined loading condition across all experiments. The same layout is used across all subsets of the dataset. To ensure consistency across all sensor modalities, every recording session follows a fixed preparation protocol.

At the start of each data collection session, the FBGS fiber calibration matrix is updated twice to minimise measurement uncertainty. First, using a self-leveling laser (Bosch Professional GLL50-20), the robot is verified to be straight along both the x and y axes, with cable pre-tensions compensating any residual backbone deformation; the calibration matrix is then updated to match this straight reference configuration. Second, the robot is bent significantly in the yz plane by actuating the $+y$ tendon, with the laser confirming that the deformation remains planar and correcting any out-of-plane component via the remaining cables; the calibration matrix is updated again from this configuration to eliminate any residual torsional offset in the fiber measurements. Despite this procedure, a residual angular offset between the fiber deformation plane and the robot bending plane is nonetheless expected, as bonding the fiber to the backbone with sub-degree angular accuracy is not achievable in practice. This offset is characterized during data collection via dedicated calibration sweeps and corrected in post-processing (Sections 2.5 and 2.6.3).

Before each experiment, the base frame is verified to be level using a circular bubble level, ensuring that the gravitational load direction is consistent with the most common modeling assumptions for tendon-driven continuum robots. The laser is then used to confirm that the robot is straight. In this configuration, the tendons are pre-tensioned to approximately 1 N each, with a maximum difference between the two cables of an antagonistic pair <0.2 N. This step minimizes the risk of cable slack, a well-known source of modeling error in tendon-driven continuum robots [18].

2.5 Recording protocol

This section describes the recordings that constitute the dataset. Once the fiber calibration procedure described in the previous section is complete, two reference recordings are performed before the main data



(a) (b)

(c) (d) (e)

Figure 3: Motion of the robot tip (xy plane) for the planar motions along x (a) and y (b) planes, the star (c), circular (d), and Lissajous (e) trajectories. Blue: slow speed ($\omega_{\max} = 1\,\mathrm{rad/s}$); red: fast speed ($\omega_{\max} = 3\,\mathrm{rad/s}$).

collection begins. The dataset is then organised into three subsets corresponding to distinct experimental conditions, each targeting a different class of robot behaviour.

Reference recordings. Two preliminary recordings are performed, stored in the `references/` folder. First, with all tendons released, a baseline recording is acquired to characterize the noise floor of all sensors in the absence of any load or motion (`released_config`). Then, with the robot verified straight using the Bosch laser, a second recording is acquired to provide an estimate of the residual alignment error in the FBGS and motion capture measurements under the known straight configuration (`straight_config`).

Quasi-static subset. The 16 recordings in `quasi_static/` are obtained by driving one pair of antagonist tendons, with maximum angular velocity of $\omega_{\max} = 0.3\,\mathrm{rad/s}$, so that inertial and damping effects are negligible and the robot reaches a sequence of near-equilibrium configurations. Bending is applied independently along $+x$, $-x$, $+y$, and $-y$, each at four commanded amplitude levels corresponding to maximum motor angles of 100, 150, 180, and 200°. Each recording sweeps the bending amplitude in small increments, producing synchronized measurements of tendon tensions, encoder angles, OptiTrack disk poses, and FBGS curvatures across a range of near-equilibrium configurations.

Dynamic motion subset. The 10 recordings in `dynamic_motion/` cover five trajectory types executed at two speeds ($\omega_{\max} = 1\,\mathrm{rad/s}$ and $\omega_{\max} = 3\,\mathrm{rad/s}$), producing a range of dynamic regimes (Fig. 3). The slow speed produces moderate dynamic effects while the fast speed amplifies inertial and damping contributions. The planar trajectories: (`plane_x` and `plane_y`), drive a single antagonist cable pair confining the motion to either the xz- or yz-plane, respectively. These recordings serve as controlled baselines: the actuation involves one degree of freedom and the expected robot shape is a planar arc. The `star` trajectory alternates between the x- and y-cable pairs and their combination in successive segments, sweeping the tip through four quadrants in a star-like pattern, introducing transitions between bending planes and richer coupling between the two actuation channels. Finally, the `circle` and `Lissajous` trajectories drive both cable pairs simultaneously with sinusoidal commands, producing continuous out-of-plane tip motion with sustained coupling between the x and y channels. The Lissajous trajectory additionally changes direction multiple times per cycle, generating more varied accelerations and richer spectral content than the circle.

For every non-planar trajectory (*i.e.* star, circular and Lissajous curves) the robot executes a calibration routine before the target motion. The routine consists of a planar sweep in the x–z plane (positive x followed by negative x), then a planar sweep in the y–z plane (negative y followed by positive y). Because aligning the FBGS fiber to the robot backbone with sub-degree accuracy is difficult in practice,

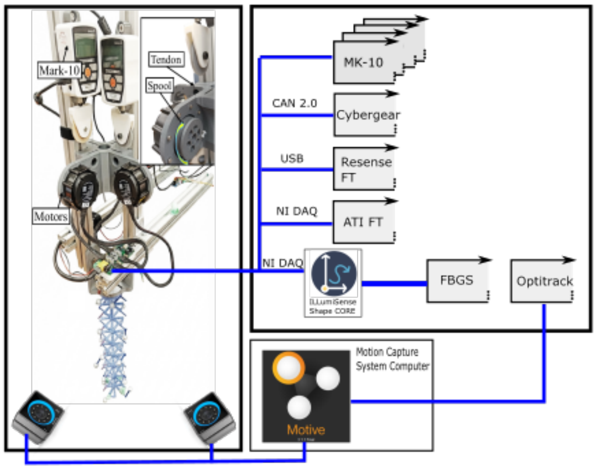


Figure 4: Representation of the hardware interfaces for streaming sensor data

a residual angular offset between the fiber deformation plane and the robot bending plane is expected. The calibration sweeps characterize this offset, allowing it to be corrected in post-processing before the non-planar data is interpreted (Section 2.6.3).

Contact motion subset. The four recordings in `contact_motion/` extend the dataset with external contact scenarios. All experiments use the instrumented contact wand (Section 2.3), so that the applied contact force vector and the 6-DoF wand pose are recorded alongside all other robot sensors. Contact experiments are restricted to the planar-y trajectory (with $\omega_{\max} = 1\,\mathrm{rad/s}$) because the geometrical constraints of the experimental setup — the thin spacer disks, the OptiTrack marker brackets, and the small tip area of the Resense sensor — make reliable and repeatable contact difficult to achieve on more complex three-dimensional motions. `push_retract` records the response of the un-actuated robot to an external tip load: the tendons are fully released and the wand is slowly pushed against the tip then retracted. `plane_y_contact_midway` is an actuated experiment in which the robot executes the trajectory while the wand is held fixed against the spacer disk located 18 cm from the base, blocking the robot's motion at that point and redistributing internal forces. `plane_y_poke_y` and `plane_y_poke_x` are dynamic perturbation experiments in which, during the robot motion, the wand delivers brief transverse pokes in the y- and x-directions respectively, exciting in-plane and out-of-plane oscillations.

2.6 Data Collection

This section describes the acquisition pipeline, synchronization strategy, and post-processing steps applied to produce the released dataset.

2.6.1 Data collection interfaces

Data acquisition is handled by a combination of manufacturer-provided software and custom scripts, depending on the sensor. Two sensors rely on dedicated proprietary software running as independent processes. The FBGS shape sensor is managed by the manufacturer's ShapeCore software, which connects to the interrogator, performs spectral peak detection, and reconstructs the three-dimensional shape in real time; the output is streamed over a local TCP connection to a custom C++ client that timestamps and buffers each frame. The OptiTrack motion capture system is managed by Motive 3.4 running on a dedicated host PC, which solves 3-D rigid-body poses at 120 Hz and streams them over a point-to-point Ethernet link using the NatNet 4.4 SDK. A custom Python client on the acquisition machine receives and timestamps each frame. All remaining sensors are read by dedicated Python processes: the Mark-10 force gauges via a serial request–response protocol (one process per gauge), the ATI force/torque sensor via a National Instruments DAQ card using the `nidaqmx` driver, the Cybergear motors via a CAN 2.0 bus accessed through a USB–CAN adapter, and the Resense contact force sensor via USB. The acquisition layout is represented in Fig. 4.

Each acquisition process runs independently and logs its data to a dedicated CSV file at the end of the recording. A hardware trigger was not used: the heterogeneous interface mix — serial, CAN, analog DAQ, TCP, and UDP — would require a dedicated trigger input on every device simultaneously, which is not supported across this sensor suite without custom electronics. All processes timestamp their readings using the operating system wall clock of the acquisition machine (`time.time()` in Python, `std::chrono::high_resolution_clock` in C++), providing a common time reference across all modalities. The motor process is launched last and waits 2 s before sending position commands, ensuring all sensors are already streaming before the actuators begin to move and every sensor captures a pre-motion baseline at the start of each recording. All acquisition scripts are released alongside the dataset.

2.6.2 Synchronization

All sensors except the FBGS share the acquisition machine's OS clock, so their timestamps are aligned to within OS scheduling jitter, estimated at 1 ms to 2 ms. The FBGS unit runs an independent internal clock and introduces an additional pipeline delay that is measured empirically during technical validation (Section 4): cross-correlation of the FBGS tip position against the OptiTrack ground truth across three dynamic trajectories yields a consistent lag of 13.4(7) ms (FBGS lags mocap), with motor–mocap offsets of −0.5 ms on average, confirming OS-clock alignment. This delay is removed in the released dataset by subtracting 13.4 ms from all FBGS timestamps in `process_data.m`. Given that the dominant dynamics in the dataset are below 20 Hz (period ≈ 50 ms), the residual sub-millisecond uncertainty after correction corresponds to a phase error well below 1% of the shortest signal period.

2.6.3 Post-processing

Post-processing aims to align all sensor measurements to a common reference frame and time base, to reduce measurement noise, and to provide uniformly sampled signals for downstream analysis. It is implemented in `process_data.m` (MATLAB), released alongside the dataset. Both raw and processed files are provided; users who require different filtering, resampling, or frame conventions should work from the raw files.

Coordinate frame alignment. Raw motion capture poses are expressed in the Motive global frame, which is arbitrary and calibration-dependent. They are re-expressed relative to a body frame coincident with the base of the robot (disk 0), making poses changes directly interpretable as robot deformation. The transformation is computed from a 120 samples-window at the beginning of each recording.

For the FBGS data, the interrogator software reconstructs the fiber shape with the arc-length parameter evolving along its internal x-axis, while the physical robot hangs along the z axis of its base frame; a rigid rotation is therefore applied to map the FBGS coordinate frame to the robot body frame. A residual rotational offset between the fiber deformation plane and the robot bending plane is additionally corrected via a singular-value decomposition performed on the tip trajectory during the initial planar calibration sweep: the principal directions of the tip motion identify the bending plane and the corresponding rotation angle θ_{FBGS}. A matching SVD is applied to the motion-capture tip trajectory, yielding θ_{mocap}, and the correction applied to the FBGS shape is a rotation of $\theta_z = \theta_{\mathrm{FBGS}} - \theta_{\mathrm{mocap}}$ about the z-axis.

Temporal alignment. All timestamps are expressed relative to the first motor command so that $t = 0$ marks the start of motion. Samples recorded before this event (sensor warm-up, static baseline) are retained in the raw files and discarded from the processed files. Users requiring pre-motion baseline data should use the raw files.

Low-pass filtering. Each signal is filtered with a zero-phase, fourth-order Butterworth low-pass filter (cutoff 20 Hz, applied via `filtfilt`). The cutoff is chosen to be well above the mechanical bandwidth of the robot under the trajectories in this dataset — the highest actuation speed corresponds to tip oscillations below 5 Hz — while attenuating sensor noise and aliasing artefacts at higher frequencies. A Butterworth filter was chosen as it is the standard choice for sensor data smoothing and its behaviour is well characterised in the literature, while zero-phase filtering eliminates phase distortion (`filtfilt`). Users interested in higher-frequency dynamics should work from the raw files.

Resampling. All filtered signals are resampled onto a uniform 100 Hz grid by linear interpolation. This rate matches the native rate of the slowest sensors (FBGS and Resense) and provides more than twofold Nyquist margin relative to the 20 Hz filter cutoff. Linear interpolation is sufficient because all signals are already band-limited by the preceding filter.

Table 3: Complete file and column documentation for each experiment folder. Subscript $n \in \{1, 2, 3, 4\}$ denotes motor index (motors 1 & 3 form the x-plane pair; 2 & 4 the y-plane pair). Subscript $k \in \{0, \ldots, 4\}$ denotes disk index from base to tip. Raw files include a header row; processed files are header-less.

File	**Column(s)**	**Unit**	**Description**
`dataMotor.csv`	`timestamp`	s	CPU wall-clock
	`phase_index`	–	Trajectory phase (0 = idle; successive integers = motion segments)
	`home`n ($n = 1$–4)	bool	Homing completed flag, motor n
	`target`n`_rad`	rad	Commanded shaft angle, motor n
	`abs_angle`n`_rad`	rad	Absolute encoder reading, motor n
	`rel_angle`n`_rad`	rad	Relative angle from home, motor n
	`velocity`n`_rad_s`	rad/s	Shaft angular velocity, motor n
	`torque`n`_Nm`	N m	Output torque, motor n
`dataMark10_*.csv`	`timestamp`	s	CPU wall-clock
(4 files: `+x`, `-x`, `+y`, `-y`)	`tension (N)`	N	Tendon tension; file suffix indicates tendon direction
`dataATIFT.csv`	`timestamp`	s	CPU wall-clock
	`Fx (N)`, `Fy (N)`, `Fz (N)`	N	Base reaction force along x, y, z
	`Tx (Nm)`, `Ty (Nm)`, `Tz (Nm)`	N m	Base reaction torque about x, y, z
`dataOptiTrack.csv`	`timestamp_s`	s	CPU wall-clock
	`disk_`k`_frame_number`	–	Motive software frame counter
	`disk_`k`_is_valid`	bool	Tracking validity (0 = occluded)
	`disk_`k`_x/y/z`	m	3-D position of disk k
	`disk_`k`_q[xyzw]`	–	Orientation quaternion, (k = 0–4 for robot disks; in subset 3 also k = 5 for the contact wand)
`dataFBGS.csv`	`Timestamp`	s	CPU wall-clock
	`Curvature_0 – _25` (26)	1/mm	Curvature at each of the 26 fiber gratings
	`Angle_0 – _25` (26)	rad	Bending angle at each grating
	`Shape_0 – _1505` (1 506)	mm	Flattened $[x, y, z]$ triplets of the reconstructed 3-D backbone shape at 1 mm arc-length resolution
`dataResenseFT.csv`	`timestamp (s)`	s	CPU wall-clock
(subset 3 only)	`Fx`, `Fy`, `Fz`	N	Contact force along x, y, z
	`Tx`, `Ty`, `Tz`	mN m	Contact torque about x, y, z
`angles.csv`	col 1: `time`	s	Time
	cols 2–5: $\theta_1, \theta_2, \theta_3, \theta_4$	rad	Motor shaft angles relative to home
`cable_tensions.csv`	col 1: `time`	s	Time
	cols 2–5: $T_{+x}, T_{+y}, T_{-x}, T_{-y}$	N	Low-pass filtered tendon tensions
`base_wrench.csv`	col 1: `time`	s	Time
	cols 2–4: F_x, F_y, F_z	N	Base reaction force
	cols 5–7: T_x, T_y, T_z	N m	Base reaction torque
`mocap_frames.csv`	col 1: `time`	s	Time
	x_k, y_k, z_k ($k = 0$–4)	m	Disk position in robot base frame
	verify$q_{x,k}, q_{y,k}, q_{z,k}, q_{w,k}$	–	Disk orientation quaternion
`fbgs_shapes.csv`	col 1: `time`	s	Time
	cols 2–175: $[x_i, y_i, z_i]$, $i = 0$–57	m	Reconstructed 3-D backbone at 1 mm resolution, aligned to robot base frame
`contact_wrench.csv`	col 1: `time`	s	Time
(subset 3 only)	cols 2–4: F_x, F_y, F_z	N	Low-pass filtered contact force
	cols 5–7: T_x, T_y, T_z	mN m	Low-pass filtered contact torque
`wand_pose.csv`	col 1: `time`	s	Time
(subset 3 only)	x, y, z	m	Contact wand position in robot base frame
	verifyq_x, q_y, q_z, q_w	–	Contact wand orientation quaternion

3 Data Records

The dataset is hosted on figshare at `https://doi.org/10.6084/m9.figshare.31990188` [19]. It is organised in four top-level directories corresponding to the three experimental subsets and the reference recordings introduced in Section 1. Table 3 documents every file type, its columns, units, and content.

3.1 File naming and outputs

Each sensor process writes one CSV file per experiment following the naming convention `data{Sensor}.csv` (e.g. `dataMotor.csv`, `dataMark10_+x.csv`, `dataATIFT.csv`, `dataOptiTrack.csv`, `dataFBGS.csv`). Column headers and units for each file are listed in Table 3. The post-processed data are saved in a `processed/` subfolder within each experiment directory, containing one file per quantity (`angles.csv`, `cable_tensions.csv`, `base_wrench.csv`, `mocap_frames.csv`, `fbgs_shapes.csv`), all resampled to 100 Hz with a shared time column. For subset 3, the `processed/` folder additionally contains `contact_wrench.csv` and `wand_pose.csv`; the corresponding raw file is `dataResenseFT.csv`. Both raw and processed data are released.

3.2 Dataset Structure

The files described above are organised in a hierarchical directory tree. At the top level, four directories separate the three experimental subsets and the reference recordings:

```
data/
  quasi_static/
    static_bend_x_100/     ... static_bend_x_200/
    static_bend_x_n100/    ... static_bend_x_n200/
    static_bend_y_100/     ... static_bend_y_200/
    static_bend_y_n100/    ... static_bend_y_n200/
  dynamic_motion/
    plane_x_slow/   plane_x_fast/
    plane_y_slow/   plane_y_fast/
    star_slow/      star_fast/
    circle_slow/    circle_fast/
    Lissajous_slow/ Lissajous_fast/
  contact_motion/
    push_retract/     plane_y_contact_midway/
    plane_y_poke_y/   plane_y_poke_x/
  references/
    released_config/
    straight_config/
```

Each experiment folder (e.g., `circle_slow/`) contains the raw sensor CSV files at the top level and a `processed/` subfolder with the filtered, resampled data:

```
<experiment>/
  dataMotor.csv        dataATIFT.csv
  dataMark10_+x.csv    dataMark10_-x.csv
  dataMark10_+y.csv    dataMark10_-y.csv
  dataFBGS.csv         dataOptiTrack.csv
  processed/
    angles.csv         cable_tensions.csv
    base_wrench.csv    base_wrench_raw.csv
    mocap_frames.csv   fbgs_shapes.csv
```

The `quasi_static/` subset contains 16 experiments corresponding to planar bends along $\pm x$ and $\pm y$ at four commanded amplitudes (maximum tendon actuation angles of 100, 150, 180, and 200°). The `dynamic_motion/` subset contains 10 experiments (5 trajectories $\times$ 2 speeds). The `contact_motion/` subset contains 5 experiments: `push_retract` (quasi-static, no actuation); and four dynamic recordings at slow speed (`plane_y_contact_midway`, `plane_y_poke_y`, `plane_y_poke_x`). The `references/` folder stores two baseline recordings: `released_config` (tendons released, robot at rest) and `straight_config` (robot held straight with laser alignment).

Table 4: Sensor noise characterisation. *Measured std* is the standard deviation computed over the reference recording with the robot at rest, characterising the random noise floor.

Sensor	**Quantity**	**Unit**	**Measured std**
Mark-10 ($+x$)	Cable tension	N	<0.02
Mark-10 ($+y$)	Cable tension	N	<0.02
Mark-10 ($-x$)	Cable tension	N	<0.02
Mark-10 ($-y$)	Cable tension	N	<0.02
Motor ($+x$)	Joint angle	rad	2.0×10^{-4}
Motor ($+y$)	Joint angle	rad	2.0×10^{-4}
Motor ($-x$)	Joint angle	rad	2.0×10^{-4}
Motor ($-y$)	Joint angle	rad	2.0×10^{-4}
ATI-FT	F_x	N	0.013
	F_y	N	0.017
	F_z	N	0.944
	T_x	N m	7.1×10^{-4}
	T_y	N m	4.3×10^{-4}
	T_z	N m	7.9×10^{-3}
Resense HEX 12	F_x	N	0.011
	F_y	N	0.0076
	F_z	N	0.013
	T_x	N m	5.1×10^{-5}
	T_y	N m	2.9×10^{-5}
	T_z	N m	2.4×10^{-5}
OptiTrack disk 0	Position (x, y, z)	mm	(0.005, 0.005, 0.003)
	Orientation (X, Y, Z)	deg	(0.0043, 0.0034, 0.0049)
OptiTrack disk 1	Position (x, y, z)	mm	(0.005, 0.005, 0.004)
	Orientation (X, Y, Z)	deg	(0.0022, 0.0029, 0.0032)
OptiTrack disk 2	Position (x, y, z)	mm	(0.007, 0.008, 0.008)
	Orientation (X, Y, Z)	deg	(0.0049, 0.0126, 0.0071)
OptiTrack disk 3	Position (x, y, z)	mm	(0.008, 0.008, 0.003)
	Orientation (X, Y, Z)	deg	(0.0034, 0.0032, 0.0049)
OptiTrack disk 4	Position (x, y, z)	mm	(0.011, 0.013, 0.005)
	Orientation (X, Y, Z)	deg	(0.0068, 0.0109, 0.0077)
FBGS	Shape noise along arc (x, y, z)	mm	(0.002, 0.194, 0.233)
	Tip position (x, y, z)	mm	(0.006, 0.476, 0.508)

4 Technical Validation

This section presents the analyses performed to verify the technical quality and internal consistency of the recorded dataset. The dataset contains two families of measurements: kinematic quantities (OptiTrack poses, FBGS strain and reconstructed shape, motor encoder angles) and force quantities (Mark10 cable tensions, ATI-FT base wrench, Resense contact force). Within each family, sensors may be directly compared. Kinematic sensors are cross-validated by comparing OptiTrack tip positions with FBGS-reconstructed shapes, and encoder-derived cable displacements with motion-capture-based geometric reconstructions. On the other hand, force measurements cannot be compared directly without a physical model. Therefore, a reduced-order dynamics model is employed as a cross-correlation tool to verify consistency, as detailed in Section 4.4.

4.1 Sensor noise baseline

A dedicated static recording (`references/released_config`) was collected with the robot hanging at rest and all tendons released. This recording provides a sensor-independent characterisation of the noise floor, decoupled from any robot motion. Table 4 reports the standard deviation of each sensor measured over the full reference recording.

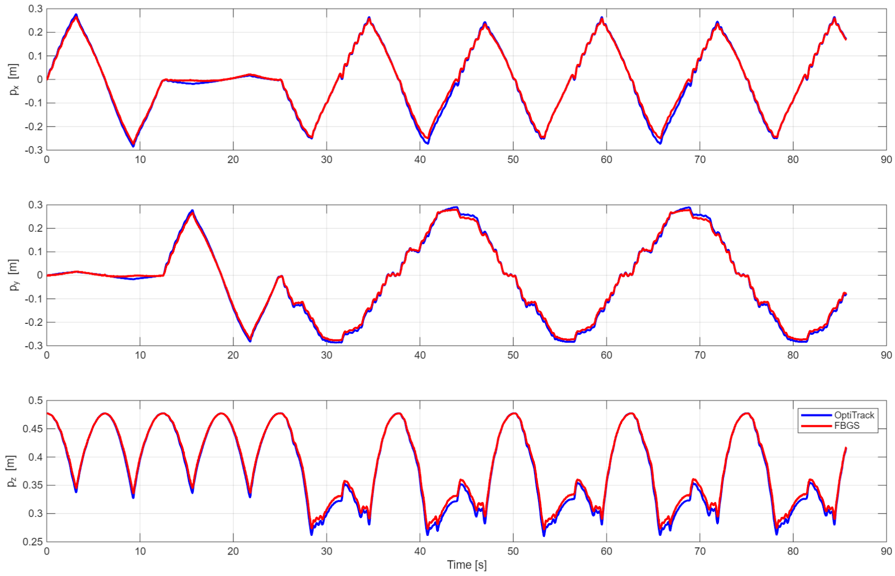


Figure 5: Trajectory of the robot tip as measured by the OptiTrack motion capture system (blue) and the FBGS (red) for the Lissajous trajectory.

Table 5: RMSE of tip position between OptiTrack and FBGS for each dynamic trajectory. Values are in millimeters; percentages in parentheses are relative to the motion range.

Trajectory	**RMSE x [mm] (%)**	**RMSE y [mm] (%)**	**RMSE z [mm] (%)**
`plane_x` slow	8.01 (1.4%)	N/A	3.62 (2.5%)
`plane_x` fast	9.88 (1.7%)	N/A	5.28 (3.4%)
`plane_y` slow	N/A	9.04 (1.6%)	3.99 (2.7%)
`plane_y` fast	N/A	10.27 (1.8%)	5.48 (3.5%)
`star` slow	7.36 (1.3%)	7.86 (1.4%)	5.34 (1.9%)
`star` fast	8.95 (1.5%)	8.75 (1.5%)	6.85 (2.3%)
`circle` slow	11.24 (1.9%)	8.98 (1.6%)	6.94 (4.0%)
`circle` fast	11.48 (2.0%)	9.67 (1.6%)	7.41 (4.0%)
`Lissajous` slow	9.21 (1.6%)	8.51 (1.5%)	6.50 (3.0%)
`Lissajous` fast	10.59 (1.8%)	9.07 (1.6%)	7.26 (3.1%)

4.2 Kinematic cross-validation

4.2.1 Comparison of recorded shape with FBGS and Motion Capture

The OptiTrack and FBGS sensors provide independent measurements of the robot shape. To verify their mutual consistency, we compare the tip position recorded by OptiTrack with the tip position reconstructed from FBGS strain data, evaluated at the same arc-length coordinate. Table 5 reports the root-mean-square error (RMSE) for all dynamic trajectories. The tip RMSE remains below 12 mm in the primary motion axes and below 8 mm in the vertical (z) direction across all trajectories and speeds, an error being 1.4–4.0% of the range of motion. The source of this error can be attributed to placement uncertainties of the OptiTrack frames and FBGS fiber, as well as the dead-reckoning error in the shape reconstruction. Nonetheless, these results confirm that the two independent kinematic measurements are mutually consistent. Figure 5 shows the overlay of these two signals for the Lissajous trajectory.

4.2.2 Validation of Cable Displacement via Motion Capture

An independent geometric estimate of the cable displacement is derived from the motion-capture data to cross-validate the encoder-based measurements recorded by the actuators. To this aim, the pose of the five measured disk frames are interpolated to locate the intermediate disks (not provided with OptiTrack

Table 6: RMSE of cable displacement between encoder and OptiTrack reconstruction for each dynamic trajectory. Values are in millimeters; percentages in parentheses are relative to the cable motion range. Cables are labeled by tendon direction ($+x$, $+y$, $-x$, $-y$).

Trajectory	$+x$ **[mm] (%)**	$+y$ **[mm] (%)**	$-x$ **[mm] (%)**	$-y$ **[mm] (%)**
`plane_x` slow	1.85 (1.7%)	N/A	3.57 (3.1%)	N/A
`plane_x` fast	2.43 (2.2%)	N/A	3.53 (3.1%)	N/A
`plane_y` slow	N/A	2.87 (2.5%)	N/A	2.44 (2.1%)
`plane_y` fast	N/A	2.77 (2.5%)	N/A	3.05 (2.6%)
`star` slow	1.41 (1.2%)	2.31 (2.0%)	1.93 (1.7%)	2.09 (1.8%)
`star` fast	1.88 (1.6%)	2.78 (2.5%)	2.20 (1.9%)	2.29 (2.0%)
`circle` slow	2.67 (2.3%)	3.74 (3.2%)	3.00 (2.6%)	3.12 (2.7%)
`circle` fast	3.15 (2.7%)	4.17 (3.6%)	3.32 (2.9%)	3.29 (2.8%)
`Lissajous` slow	1.22 (1.1%)	2.38 (2.0%)	1.73 (1.5%)	2.15 (1.8%)
`Lissajous` fast	1.93 (1.7%)	2.58 (2.2%)	2.02 (1.7%)	2.18 (1.9%)

frames). Positions are interpolated with a shape-preserving piecewise-cubic Hermite (PCHIP) scheme, while orientations are interpolated by converting each rotation matrix to a unit quaternion, applying PCHIP independently to each quaternion component, and normalizing the result to the unit sphere. The consistency of the quaternion hemisphere is enforced between successive disks to avoid sign-flip artifacts. In each interpolated frame, the absolute position of every cable is computed by applying the local cable offset d_c on each disk pose. The total length of the cable ℓ_c^d, where the suffix indicate the relation with the disks pose, is then approximated as the piecewise-linear sum of Euclidean distances between consecutive attachment points along the interpolated backbone. Cable displacement is defined as the change in length relative to the straight (pre-motion) configuration, estimated by averaging the first $N_{\text{ref}} = 10$ frames:

$$\Delta\ell_c^d(t) \;=\; \ell_c^d(t) - \bar{\ell}_c^{d,\,\text{ref}}, \qquad \bar{\ell}_c^{d,\,\text{ref}} = \frac{1}{N_{\text{ref}}} \sum_{k=1}^{N_{\text{ref}}} \ell_c^d(t_k). \tag{1}$$

On the other hand, an independent estimate is obtained from the motor encoders (see Section 2.2). The two estimates are compared directly in the time domain. Table 6 reports the RMSE between encoder-based and motion-capture-based cable displacement for all dynamic trajectories. The cable RMSE remains below 4.2 mm across all trajectories, corresponding to less than 3.6% of the motion range, confirming that the encoder recordings and the motion-capture poses are geometrically consistent. Agreement between them constitutes a geometric consistency check, confirming that the encoder records and the motion-capture poses are mutually consistent and that the assumed cable-routing geometry is correct.

4.3 Consistency of contact measurements

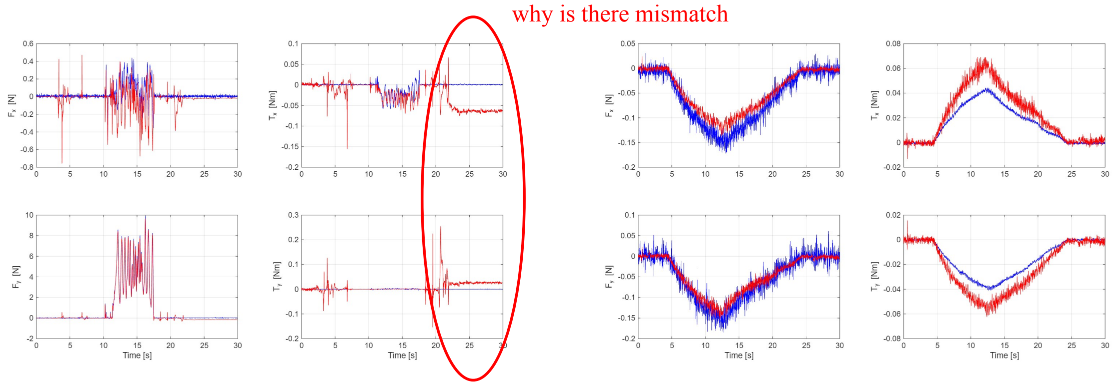
The contact subset involves two quantities that require independent verification: the position of the contact sensor and the contact wrench it measures. These are first checked in isolation and then in combination.

The contact wand carries a dedicated OptiTrack frame. To verify that the sensor position is correctly recorded throughout the robot's deformation range, the Euclidean distance between the tracked wand position and the centre of the robot tip disk was computed over the full `push_retract` recording. The mean distance was ≈38 mm with negligible standard deviation, consistent with the known physical offset between the Resense sensor origin and the disk centre and confirming accurate position tracking across all configurations.

The Resense HEX 12 contact sensor was validated by comparing its measured wrench against the ATI-FT base wrench via rigid-body transformation. In both tests, all tendons were released so that the only external load on the robot is the contact force applied by the wand. In the first test (`touch_base`), the wand was pressed against the robot base disk along the y-axis of the base OptiTrack frame, keeping the lever arm between the contact point and the ATI-FT sensing origin short and well-defined. The Resense

Need to increase the font size of these plots. What do blue and red mean?

why is there mismatch



(a) (b)

Figure 6: Comparison of the measured forces (F_x and F_y) and torques (T_x and T_y) from the measurement of the Resense to the ATI FT sensors for the touch_base (a) and push_retract (b) recordings.

Table 7: RMSE between GVS-simulated and ATI-FT-measured base torque for each dynamic trajectory. Values are absolute (N m) with the percentage of the signal range in parentheses. The two components T_x and T_y correspond to the bending moments about the x and y axes at the robot base.

Trajectory	**Speed**	**RMSE T_x [N m] (%)**	**RMSE T_y [N m] (%)**
`plane_x`	slow	N/A	0.0162 (7.8%)
	fast	N/A	0.0474 (19.4%)
`plane_y`	slow	0.0160 (7.6%)	N/A
	fast	0.0417 (17.0%)	N/A
`star`	slow	0.0141 (6.4%)	0.0138 (6.2%)
	fast	0.0342 (13.9%)	0.0362 (15.2%)
`circle`	slow	0.0155 (6.9%)	0.0163 (7.2%)
	fast	0.0357 (13.7%)	0.0391 (15.2%)
`Lissajous`	slow	0.0148 (6.3%)	0.0147 (7.0%)
	fast	0.0389 (17.6%)	0.0387 (15.8%)

wrench was mapped into the ATI-FT coordinate frame via the adjoint map

$$\mathcal{W}^{\text{base}}_{\text{contact}} = -\text{Ad}_{g_{base,contact}}\, \mathcal{W}^{\text{wand}}_{\text{Resense}}, \tag{2}$$

where $g_{base,contact} = g_{base}^{-1} g_{contact}$ is the OptiTrack-measured relative pose between the base disk and the wand, and the minus sign accounts for the action-reaction principle. The adjoint transports a wrench between frames without assuming rigidity of the connecting body, so Eq. (2) remains valid as the continuum robot deforms. Figure 6a shows close agreement between the transformed Resense wrench and the ATI-FT measurement, confirming that the Resense calibration and the geometric transformation are mutually consistent.

In the second test (`push_retract`), the wand was used to push the robot tip through a quasi-static deformation, combining both checks: the position tracking established above feeds the full pose chain from wand to base (lever arm up to 0.48 m) into the same transformation. Figure 6b shows good agreement in the force components across the full deformation range. The torque components exhibit an offset that grows with robot deformation; this is expected: as the robot bends, a growing fraction of its weight is registered by the Resense but not at the ATI-FT. The force agreement across the full deformation range confirms the consistency of both force sensors and the pose-based wrench transformation.

4.4 Consistency of force measures

Cable tensions and base wrench measurements cannot be directly compared without a physical model relating tendon forces to base reactions. As an auxiliary consistency check, we employ a forward dynamics simulation based on the Geometric Variable Strain (GVS) approach [7], using the measured tendon tensions as input to predict the base torque and comparing the result against the ATI-FT measurements. This check is model-dependent: the agreement reflects the joint quality of the data and the accuracy of the

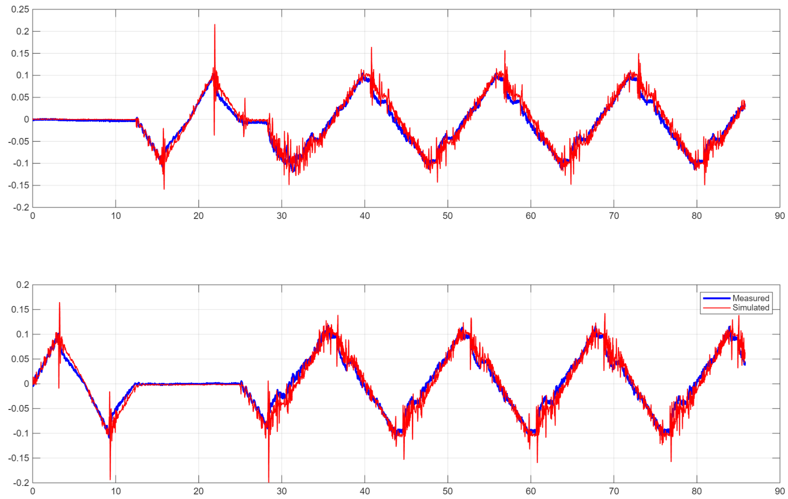


Figure 7: Comparison of the simulated torque at the base (red) with the measured torque by the (unfiltered) ATI FT sensor (blue) using the circular motion.

characterised mechanical parameters, and should not be interpreted as an independent sensor validation. The GVS simulation code and its outputs are not part of the released dataset. The equivalent bending rigidity EI of the backbone was estimated from the quasi-static recordings; inertial parameters were derived from CAD without further identification.

Figure 7 compares the simulated and ATI-FT-measured base torque for the circular trajectory; the overall trend is well captured, with residual discrepancies (e.g. torque spikes at fast speed) attributed to the approximate distributed inertia parameters. Table 7 reports the RMSE across all dynamic trajectories for the two base-torque components T_x and T_y; the component not excited by a given planar trajectory (e.g. T_x for `plane_x`) is omitted. For slow trajectories, RMSE remains below 0.02 N m ($<$ 8% of the signal range); at fast speed it increases to $\sim$0.04 N m ($\sim$15–20%), consistent with the greater contribution of inertial terms that are only approximately characterized. The qualitative agreement across all trajectories supports the internal coherence of the force and kinematic recordings, noting that residual discrepancies reflect the approximate nature of the model parameters rather than inconsistencies in the data.

4.5 Temporal synchronization

All sensors share a single operating-system clock (see Section 2.6.2), bounding the theoretical inter-stream error to OS scheduling jitter of 1–2 ms. Temporal alignment was verified empirically via pairwise normalized cross-correlation $r \in [-1, 1]$, , where $|r| = 1$ indicates a perfectly linear relationship between two signals — positive for co-varying pairs, negative for kinematically anti-correlated ones — and $|r| \approx 0$ indicates insufficient shared motion content for reliable lag estimation. Three fast trajectories (*plane_x_fast*, *plane_y_fast*, and *circle_fast*) were selected because their high-amplitude, spectrally rich motion produces sharp cross-correlation peaks; for motor–sensor pairs, only the axis predominantly excited by each trajectory is used to avoid the kinematic coupling artifacts that arise on passive axes.

Cross-correlation between OptiTrack and FBGS tip-position signals revealed a systematic pipeline delay: on the circular trajectory, where all three Cartesian axes are fully excited, $\Delta_{\mathrm{OF}} = -13.4 \pm 0.7$ ms ($|r| \approx 1.00$ on all three axes, FBGS lagging). Direct motor–FBGS cross-correlation on the same trajectory gives $\Delta_{\mathrm{MF}} = -13.1$ ms and -14.6 ms on the x- and y-axes respectively ($|r| \approx 1.00$). Triangle closure then confirms internal consistency: the identity $\Delta_{\mathrm{MF}} = \Delta_{\mathrm{MO}} + \Delta_{\mathrm{OF}}$ (predicted -13.9 ms) holds to within 1.2 ms and 1.9 ms on the two axes, well within the estimation uncertainty.

Indirect evidence of alignment is provided by the RMSE values in Tables 5 and 6: a residual timing offset would inflate RMSE in proportion to signal slope, yet tip-position RMSE between OptiTrack and FBGS remains below 12 mm (1.9% of motion range) and cable-displacement RMSE remains below 4.2 mm (3.6%), consistent with geometric and calibration uncertainties.

5 Usage Notes

Getting started. The dataset is available for download from figshare (`https://doi.org/10.6084/m9.figshare.31990188`). After downloading, users should select the experimental subset of interest (quasi-static, dynamic motion or contact) and, within it, the trajectory type they wish to analyze. All subsets expose the same file structure under their `processed/` subfolder, so downstream code does not need to distinguish between them: each experiment folder contains one CSV per sensor stream with a common time axis, pre-filtered signals, and aligned timestamps.

Users who wish to regenerate the processed files from the raw CSVs, or to apply alternative filtering parameters, should run `process_data.m` on the desired experiment folder. The `references/straight_config/` recording should be loaded first to compute the OptiTrack frame correction, while `references/released_config/` provides a static noise baseline for computing per-experiment sensor bias.

Recommended subset per task. The three subsets address complementary research tasks and can be used independently or in combination. Researchers working on mechanical parameter identification should use `subset_1` (quasi-static): each recording provides steady-state disk poses from OptiTrack alongside simultaneous Mark-10 cable tensions and motor encoder readings at a known configuration, enabling estimation of backbone bending rigidity EI and cable axial stiffness. Researchers working on state estimation, system identification, or benchmarking of control and prediction methods should use `subset_2` (dynamic), which provides full multi-sensor time series across a range of trajectories and actuation speeds. Researchers working on contact dynamics or wrench estimation should use `subset_3` with contact force and pose measurements.

Sensor-specific guidance. *FBGS shape data.* The FBGS fiber cannot be bonded to the backbone with sub-degree angular accuracy; a residual rotational offset between the fiber deformation plane and the robot bending plane is corrected in post-processing using the planar calibration sweeps (see Section 2.5). Users working with the raw `dataFBGS.csv` files must apply this correction themselves before interpreting bending angles or comparing FBGS shape with OptiTrack poses. In addition, the shape reconstruction algorithm integrated in the proprietary FBGS acquisition software accumulates a dead-reckoning error along the fiber arc: small per-segment orientation errors integrate into a positional drift that grows towards the tip.

ATI-FT base wrench. The ATI Mini40 exhibits a high noise floor on the F_z axis ($\approx 0.94\,\mathrm{N}$). Users should restrict analyses requiring sub-newton axial force resolution to F_x and F_y; T_x and T_y are the most reliable quantities for bending moment analysis.

Motor encoder data. Shaft velocity and output torque are included in `dataMotor.csv` but have not been independently validated; users relying on these quantities should treat them as indicative.

Software requirements. The raw and processed data are stored as plain CSV files and can be loaded in any programming language. The post-processing pipeline (`process_data.m`) was developed and tested in MATLAB R2025b, using only the built-in functions `butter`, `filtfilt`, and `interp1` from the Signal Processing Toolbox.

Limitations of applicability. This dataset is not suitable for all continuum robotics research tasks, and users should be aware of the following scope constraints before applying it. All processed signals are low-pass filtered at 20 Hz and resampled to 100 Hz; the dataset is therefore not appropriate for studying dynamics or contact transients above 20 Hz. The dataset covers a single robot platform with a fixed morphology, backbone material, and mounting configuration; results obtained on this platform may not generalise directly to robots with different backbone stiffness, tendon routing, or aspect ratios, and such generalisation should be validated separately. Contact scenarios are restricted to interactions with a rigid instrumented wand along the planar-y trajectory; soft, distributed, or multi-point contact is not represented.

6 Data Availability

The dataset described in this work is publicly available on figshare at `https://doi.org/10.6084/m9.figshare.31990188` [19]. The repository contains all raw and processed sensor recordings organized as described in the Data Records section (Section 3).

7 Code Availability

8 Author Contributions

9 Competing Interests

The authors declare no competing interests.

10 Funding

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